coordinate system complications, including mixed data sources using different reference systems, incomplete transformation documentation, and legacy systems with poorly defined coordinate frameworks. Field survey data might use local construction coordinates, while satellite imagery uses geographic coordinates and facility drawings reference an arbitrary site grid.

When we work with local coordinates in a floorplan or blueprint, we specify an origin point (0,0) and everything in the floorplan has local coordinates relative to that origin. To position this floorplan on a real-world map using global projected coordinates like the Universal Transverse Mercator (UTM) projection, we need to transform each point's coordinates. This transformation from 2D local coordinates to projected global coordinates involves two operations. First, if the floorplan is not oriented towards true North but rotated towards a 'site North', the coordinates must be rotated by the offset angle . Then, they must be translated by the offset from the target reference system's origin. To transform local 2D coordinates to UTM coordinates, a 3x3 matrix can be applied where (tx, ty) are the UTM coordinates of the origin point (easting, northing).

$$\begin{bmatrix} cos(\theta) & -sin(\theta) & tx \\ sin(\theta) & cos(\theta) & ty \\ 0 & 0 & 1 \end{bmatrix}$$

Listing 2.7 shows how we can perform this coordinate transformation in code by using matrix multiplication.